APPENDIX

Historical Perspective

The major historical events for the development of Biotechnology, as an interdisciplinary field with multidisciplinary applications are listed below:

Before Common Era

6000 BC – 3000 BC – Bread making, fermentation of fruit juices and plant exudates to produce alcoholic beverages using yeast.

Pre – 20th Century

1770 – Antoine Lavoisier gave chemical **basis of alcoholic fermentation.**

1798 – Edward Jenner uses **first viral vaccine** to inoculate a child from smallpox.

1838 – **Protein** discovered, named and recorded by Gerardus Johannes Mulder and Jons Jacob Berzelius.

1871 – Ernst Hoppe, Seyler discovered **enzyme invertase**, which is still used for making artificial sweeteners.

1876 – Louis Pasteur identified **role of microorganisms in fermentation**.

20th Century

1919 – The **term biotechnology** was coined by Karl Ereky

1928 – **Discovery of Penicillin** by Alexander Fleming

1941 – Experiment with *Neurospora crassa* resulting in **one gene one enzyme hypothesis** by George Beadle and Edward Tatum.

1944 – Identification of **DNA as the genetic material** Avery–MacLeod–McCarty

1953 – Discovery of **double helix structure of DNA** by James Watson and Francis Crick.

1972 – Discovery of **Restriction enzymes** by Arber, Smith and Nathans.

1973 – Fragmentation of DNA-combined with Plasmid DNA, **r-DNA technology** - Genetic engineering -Modified gene by Stanley Cohen, Annie Chang, Robert Helling and Herbert Boyer.

1975 – Production of **Monoclonal antibodies** by Kohler and Milstein

1976 – Sanger and Gilbert developed **techniques to sequence DNA**

1978 – Production of **human insulin** in E.Coli

1979 – **Development of Artificial gene** – functioning within the living cells by H.G. Khorana

1982 – U.S approved **humulin** (human insulin) the first pharmaceutical product of rDNA technology, for human use.

1983 – Use of **Ti plasmids** to genetically transform plants

1986 – Development of **Polymerase Chain Reaction (PCR)** technology by Kary Mullis.

1987 – Gene transfer by **biolistic transformation**

1992 – First chromosomes of yeast is sequenced

1994 – U.S approved the first **Genetically Modified food: Flavr Savr tomato.**

1997 – The **first transgenic animal**, mammalian sheep, Dolly developed by **nuclear cloning** by Ian Wilmet.

2000 – **First plant Genome** of *Arabidopsis thaliana* sequenced

21st Century

2001 – Human genome Project creates a **draft of the human genome sequence**.

2002 – **First crop plant genome sequenced** in *Oryza sativa*

2003 – **Human genome project is completed**, providing information on the locations and sequence of human genes on all 46 chromosomes.

2010 – Sir Robert G. Edwards developed ***in vitro* fertilization in animal**.

2016 – Stem cells injected into stroke patients re-enable patient to walk – **Stem cell therapy**

2017 – **Blood stem cells** grown in lab.

2018 – James Allison and TasukuHonjo **discovered protein found in immune cells.** This found a new role in cancer therapy.

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